TRAINER KEY. DO NOT DISTRIBUTE. Day 1, Session 1: What do the numbers actually say?

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Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Source: the regression table on the printed worksheet (before GreenWaste 1,432 AED, interval 1,418 to 1,446; after minus before -669 AED, interval -684 to -654; $p < 0.001$). Rule (invented for the course): at least 1,000 AED annual saving per business. The observed fall (669) and its interval (654 to 684) are below 1,000. This does not show that the programme has no effect.

Worksheet (2 pages)

1 Mean. Find the mean. Explain why it does not describe every business.

> 100 tonnes / 10 businesses = **10 tonnes**. Nine of the ten values are below the mean; one business (46 tonnes) pulls it up. It is not the value for every business, and the ten are a teaching selection, not representative of the city.

2 Coefficient. Circle the change row. Write its number, units and meaning of the sign.

> The "After minus before" row: **-669 AED** per business per year. Negative = lower annual waste management cost 12 months after than before. The coefficient is a named model result, not automatically a treatment effect. (The "Before" row, 1,432, is the starting mean.)

3 Treatment effect. Does the after/before change establish what GreenWaste caused? Explain the missing comparison.

> No. The change may include other things that changed over time (prices, weather, other programmes). We need a credible comparison showing what would have happened without GreenWaste, e.g. comparison businesses that did not take part.

4 Confidence interval. Box the interval for the change. What does it describe? Does it show the range for individual businesses?

> Box -684 to -654 AED. It describes uncertainty about the estimated mean change, under the model assumptions. It does not show the range of individual businesses' savings and does not cover all bias or establish cause.

5 P value. Find the change row p-value. Does it give the probability that GreenWaste works or establish cause?

> $p < 0.001$. It is evidence against the tested "no mean change" model under its assumptions. It is not the probability that GreenWaste works or that the result is due to chance, and it establishes neither cause nor a large enough saving.

6 Read and repair: "The tiny p-value proves GreenWaste caused enough saving to expand."

> Example repair: "Participants' mean annual cost fell by 669 AED (interval 654 to 684) 12 months after, and the tiny p-value shows that this is unlikely to be exactly zero. It does not show that GreenWaste caused the fall, and the fall and its interval are below our 1,000 AED rule, so it does not support expanding on this evidence alone."

My independent reading and next question

> Require: the named row, signed coefficient (-669), units (AED per business per year), interval (-684 to -654) and the comparison (the same participants before and after, no comparison group). Question example: "What happened to comparison businesses that did not receive GreenWaste?" Give brief feedback in Session 2 on any missed term.

AI exercise: read the regression table with an AI (about 15 minutes)

Part A: Mark the response. Prompt: "Explain this table and advise whether to expand GreenWaste."

> The city data show a substantial improvement in participant businesses' annual waste management cost: the mean fell from 1,432 AED before GreenWaste to 763 AED twelve months later. The reported change is -669 AED, with a 95% confidence interval of -684 to -654 AED and $p < 0.001$. This is strong statistical evidence against a zero mean change.
>
> Because the same businesses were measured twice, their starting differences are accounted for, ~~so the programme can be credited with this reduction.~~ ~~The narrow interval suggests the finding is robust, and most individual businesses can expect savings in this range.~~
>
> The recorded reduction is below our rule of at least a 1,000 AED reduction, ~~but the very small p-value makes a compelling case for expansion.~~ A phased rollout with monitoring would be prudent to confirm the result in other locations. *(Plausible caveat, left unmarked: it does not repair the unsupported claims.)*

Which claim would you correct first, and what would you write instead?

> Example: correct "the programme can be credited with this reduction": "A before-and-after change does not show what GreenWaste caused; we need a comparison with businesses that did not take part." Also acceptable: the "compelling case for expansion" (the reduction and interval are below 1,000 AED, and a small p-value does not show size or cause), or "most individual businesses can expect savings in this range" (the interval is about the mean).

Part B (a) Which claim improved? (b) What improved compared with Part A? (c) Which claim still needs checking against the table?

> Look for: no longer credits the programme with the fall; says the interval describes the mean, not individual businesses; notes the reduction and interval are below 1,000; asks about comparison businesses. Still to check: every number against the table, any claim about cause or "robust", and that the rule is stated correctly.

Prepared response (a) Which sentences does the regression table support? (b) What additional evidence would answer the clarifying question at the start?

> (a) The sentences giving the -669 mean change and the -684 to -654 interval, that the interval describes the mean change, that the p-value starts from a zero-change assumption, and that the recorded reduction and its interval are below 1,000. The claims that a before-and-after comparison does not show cause, and that it does not show that GreenWaste's causal effect is below the rule, are statements of what the table cannot show. (b) Outcomes for comparison businesses that did not receive GreenWaste (a credible no-programme comparison), and full programme costs.